Yifan Guo

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GitHub: github.com/Ricardo-Guo | Portfolio: Ricardo-Guo.github.io

EDUCATION

Southern University of Science and Technology (SUSTech) — Shenzhen, China
B.S. in Robotics, Expected Jun. 2027
GPA: 3.77/4.00; Average: 90.87/100
Relevant Coursework: Machine Learning, Control Engineering, Robot Modeling and Control, Mechatronics

PATENT APPLICATION

Metamorphic Aircraft, Chinese Invention Patent Application No. 2026102057666; co-inventor, 3rd listed; accepted by CNIPA.

SELECTED HONORS, AWARDS & GRANTS

- National-level Funded Project, College Student Innovation and Entrepreneurship Training Program, 2026.
- National Third Prize, National College Student Mechanical Innovation Competition, 2026.
- Provincial First Prize, National College Student Mechanical Innovation Competition, Bionic Butterfly Track, 2026.
- Provincial First Prize, Guangdong Engineering Maker Competition; team ranked 2nd overall in category, 2025.
- SUSTech Climbing Program Research Grant, RMB 20,000, 2025.

RESEARCH EXPERIENCE

Vision–LiDAR Fusion for 3D Target Localization and Motion Estimation May 2026 – Present

- Built a vision–LiDAR fusion pipeline combining YOLO detection, LiDAR point-cloud projection, and 2D–3D association for spherical-target 3D localization and velocity estimation.
- Hardware-synchronized a global-shutter camera with Livox LiDAR; implemented point-cloud projection, clustering, and image–point-cloud association, and tested the pipeline on NVIDIA Jetson.
- Validated against motion-capture ground truth, with 95% of detections within 5 m showing 3D localization error below 10 cm.

Vision-Based Dexterous-Hand Grasping with Reinforcement Learning Nov. 2025 – Present

- Collaborated with four teammates to propose a five-finger dexterous-hand structure with a reconfigurable palm, and built a UniDexGrasp++-based reinforcement-learning grasp-training pipeline.
- Built an Isaac Gym simulation environment, processed RGB-D observations, and back-projected depth maps into point-cloud inputs for grasp-policy training.
- Debugged the Arduino multi-motor control interface and deployed the trained policy on Jetson Nano Super for physical-system validation; project received national-level innovation-program support in 2026.

Metamorphic-Structure Reconfigurable Aircraft Sep. 2024 – Oct. 2025

- Participated in designing a reconfigurable aircraft for long-endurance, low-altitude exploration; built control and electromechanical actuation systems for airship/fixed-wing configuration switching.
- Built a Gazebo/ROS 2 simulation environment, debugged Pixhawk–QGroundControl communication, and implemented STM32/Arduino-based motor and servo control.
- Completed a functional prototype and ground configuration-switching demonstration; received RMB 20,000 in SUSTech research funding.

TECHNICAL SKILLS

Programming	C++, Python, Linux/Ubuntu, Git, Docker, PyTorch
Robotics Systems	ROS 2, Gazebo, Isaac Gym, Pixhawk, QGroundControl, STM32, Arduino, NVIDIA Jetson
Perception	OpenCV, YOLO, RGB-D processing, point-cloud processing, LiDAR–camera calibration, 3D localization
Learning & Control	Reinforcement learning, robotic grasping, embedded motor control
Design	SolidWorks